Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Construct a relational model for a Library Management System and explain how views can be used to simplify data access for librarians without exposing underlying schema details.	**2**	(CO1, L3)
2)	Describe the key steps involved in the requirement elicitation process and explain their importance in software development.	**2**	(CO2, L2)
3)	Analyze the importance of using Data Definition Language (DDL) and Data Manipulation Language (DML) queries in database project construction.	**2**	(CO3, L4)
4)	Examine the steps involved in the quality assurance process during software development and differentiate between the roles of a QA team and a quality engineer.	**2**	(CO1, L4)
5)	Construct a scenario where you need to elicit quality requirements for a new software project and explain how you would determine the potential impact on software performance.	**2**	(CO2, L3)

Section B **5*4=20**

6)	Construct an example database schema using DDL commands for a library management system and explain how you would update this schema with new tables to include inter-library loan functionality. Provide SQL code snippets for both creating initial tables and adding the necessary foreign keys.	**5**	(CO3, L3)
7)	Construct a relational database model for a Library Management System. Include at least three entity types with their attributes and explain how relationships between these entities can be represented in the model. Demonstrate this by creating a view that lists all books borrowed by a specific member.	**5**	(CO1, L3)

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8) a)	Design a relational model for a Library Management System using at least three entities. Explain how these entities are related and demonstrate this relationship through an example. Additionally, create a view that combines information from two of your tables to provide a summary of available books by genre.	**5**	(CO2, L3)
	OR		
b)	Determine and explain the significance of each layer in the software engineering process, using an example to illustrate how data flows through these layers.	**5**	(CO2, L3)
9)	Construct a simple database project using DDL and DML queries to manage an employee management system. Explain your schema design choices for this system.	**5**	(CO3, L3)

Section C **10*2=20**

10)	Discuss the process and importance of requirement elicitation for developing an ethical software product. In your answer, explain how potential issues can arise during this phase and suggest strategies to mitigate them. Additionally, describe the role of quality assurance (QA) in ensuring the quality of requirements throughout the development lifecycle.	**10**	(CO1, L2)
11) a)	Examine the relational model by discussing its core components such as entities, relationships, attributes, and keys. Further, analyze how views enhance data management in a database system, illustrating your answer through examples of creating, inserting into, and deleting from a view. Lastly, differentiate between an entity set and an attribute within this context.	**10**	(CO2, L4)
	OR		
b)	Examine the key challenges faced during the requirement elicitation phase and analyze how these challenges can affect the quality of software development projects. Additionally, discuss potential strategies to mitigate these issues.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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